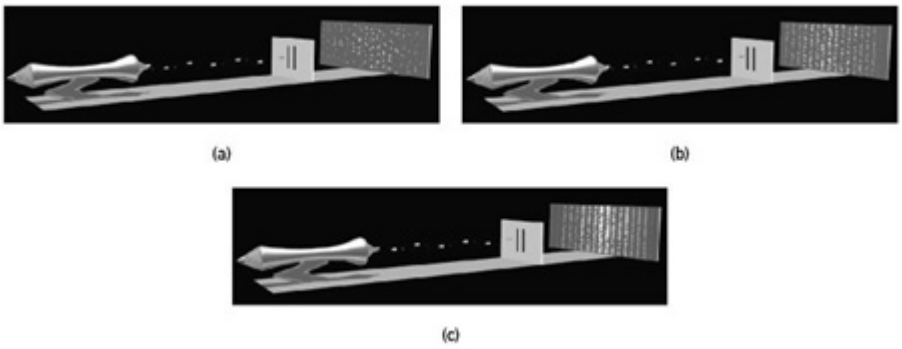


Figure 4.4 Electrons fired one by one toward slits build up an interference pattern dot by dot. In **(**a**)–(**c**)** we illustrate the pattern forming over time.

Probability and the Laws of Physics

If an individual electron is also a wave, what is it that is waving? Erwin Schrödinger weighed in with the first guess: maybe the stuff of which electrons are made can be smeared out in space and it's this smeared electron essence that does the waving. An electron particle, from this point of view, would be a sharp spike in an electron mist. It was quickly realized, though, that this suggestion couldn't be correct because even a sharply spiked wave shape—such as a giant tidal wave—ultimately spreads out. And if the spiked electron wave were to spread we would expect to find part of a single electron's electric charge over here or part of its mass over there. But we never do. When we locate an electron, we always find all of its mass and all of its charge concentrated in one tiny, pointlike region. In 1927, Max Born put forward a different suggestion, one that turned out to be the decisive step that forced physics to enter a radically new realm. The wave, he claimed, is not a smeared-out electron, nor is it anything ever previously encountered in science. The wave, Born proposed, is a *proba*bility wave.

To understand what this means, picture a snapshot of a water wave that shows regions of high intensity (near the peaks and troughs) and regions of low intensity (near the flatter transition regions between peaks and troughs). The higher the intensity, the greater the potential the water wave has for exerting force on nearby ships or on coastline structures. The probability waves envisioned by Born also have regions of high and